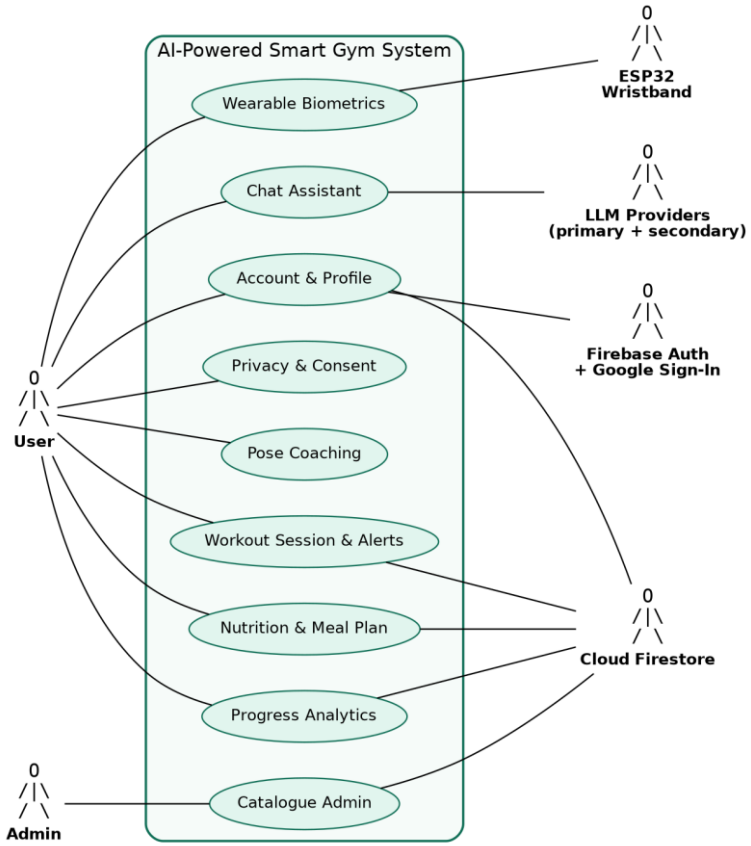
AI-Powered Smart Gym System

Use Cases and Use Case Diagrams - derived from the Requirements Engineering document. 51 use cases across 10 subsystems; each is mapped to the FR / UR / NFR IDs it realises.

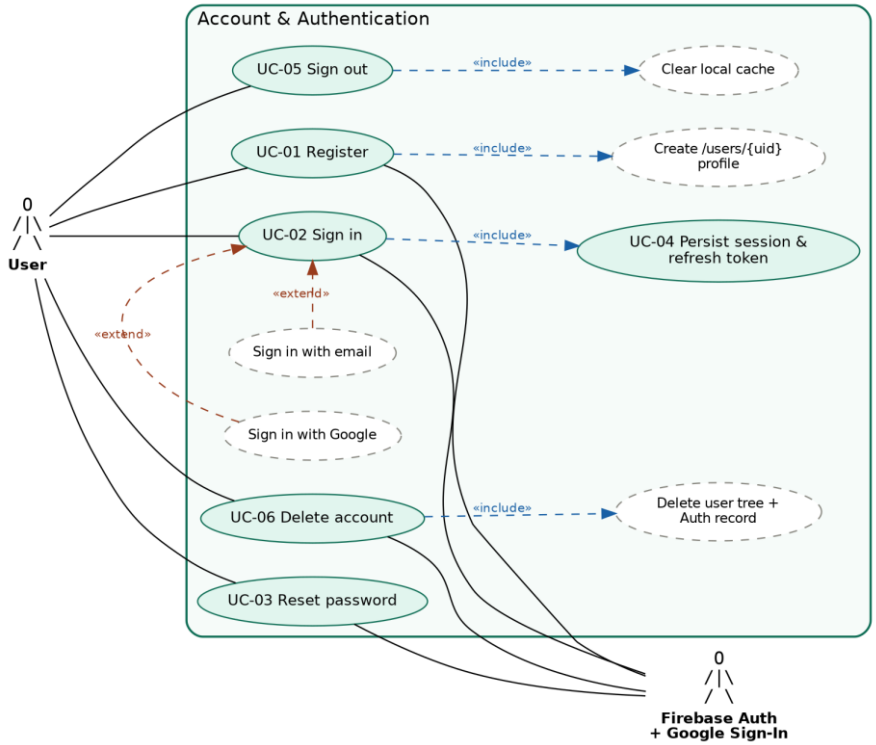
Legend: ellipses = use cases (green = numbered UC, dashed white = supporting step); stick figures = actors; blue dashed arrow = «include»; orange dashed arrow = «extend» (extension points to the base use case).

0. System context & actors



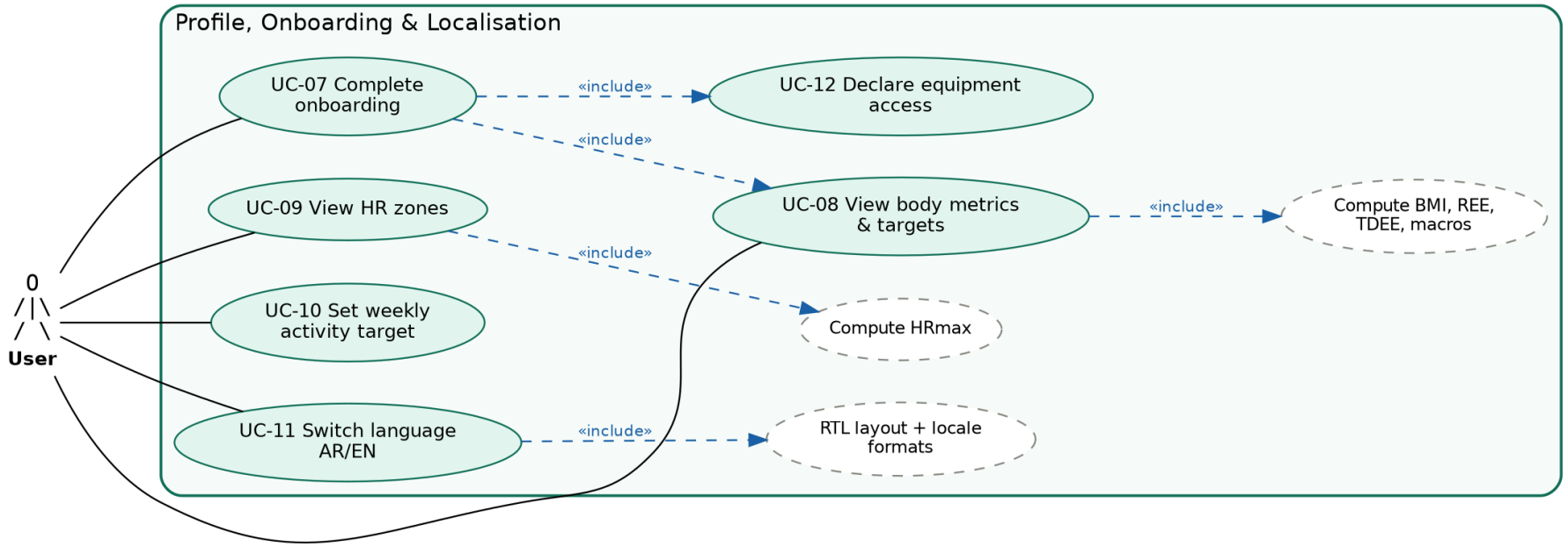
Actors: User (all four classes share the same use cases; the class only changes emphasis), Admin, and external systems Wristband, Firebase Auth, Firestore, LLM providers. The camera and BLE stack are platform services, not actors.

1. Account & authentication (FR-AUTH)



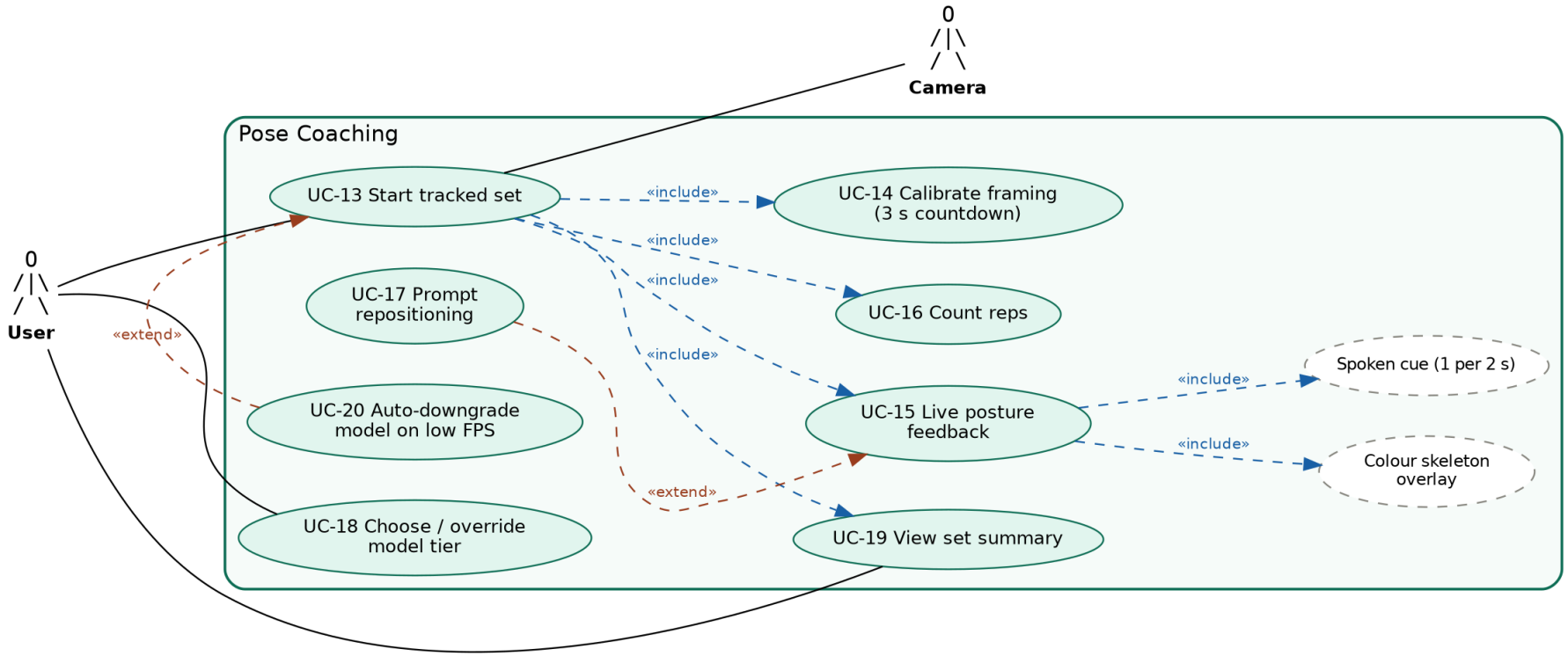
ID	Use case	Actor	Reqs
UC-01	Register with email + password; profile document auto-created	User, Firebase Auth	FR-AUTH-01, 04
UC-02	Sign in (email/password or Google; same UID if email matches)	User, Firebase Auth	FR-AUTH-02
UC-03	Reset password (link sent without revealing if email exists)	User	FR-AUTH-03
UC-04	Persist session and silently refresh token	System	FR-AUTH-05
UC-05	Sign out and wipe PII, biometrics and chat from local cache	User	FR-AUTH-06
UC-06	Permanently delete account and data	User	FR-AUTH-07, NFR-LEG-02

2. Profile, onboarding & goals (FR-PROF, FR-LOC)



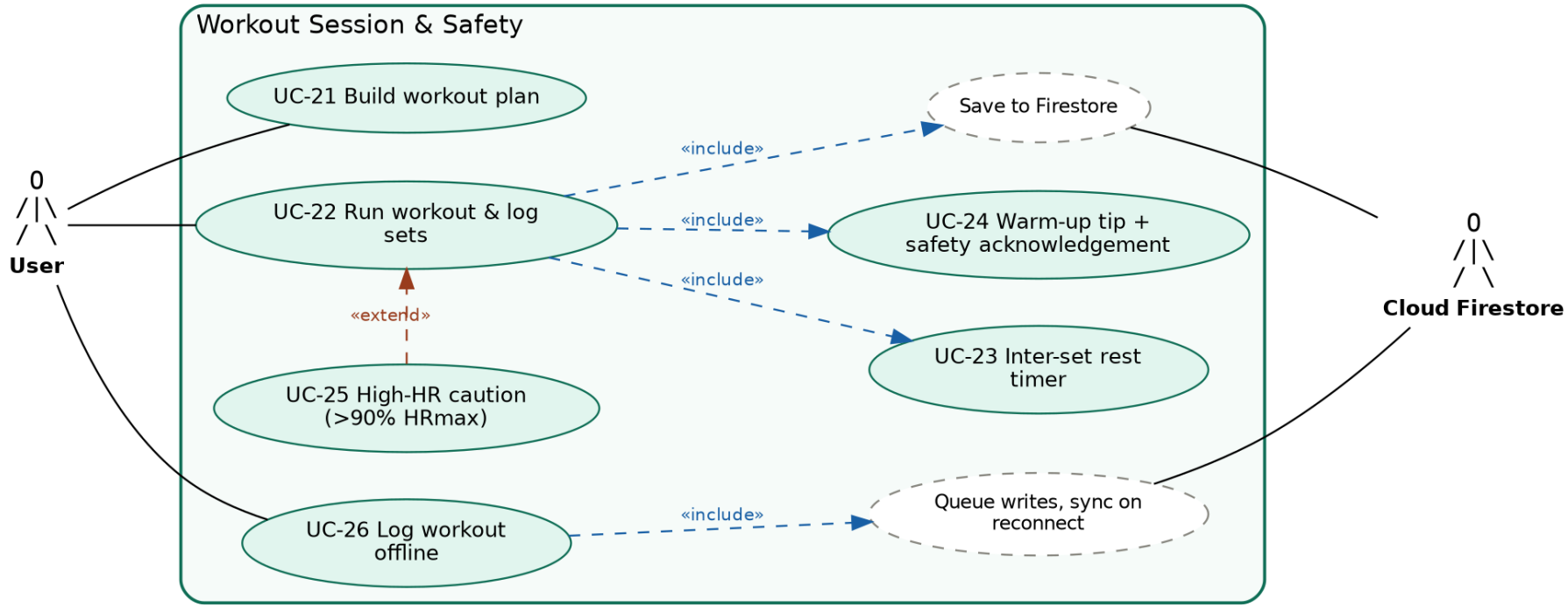
ID	Use case	Actor	Reqs
UC-07	Complete onboarding (8 attributes, ≤ 8 screens)	User	FR-PROF-01
UC-08	View/update BMI, REE (Mifflin-St Jeor), TDEE and macro targets	User	FR-PROF-02, 03, 04
UC-09	View HRmax and five intensity zones	User	FR-PROF-05
UC-10	View/override default weekly activity target	User	FR-PROF-06, NFR-LEG-06
UC-11	Change interface language without restart (full RTL mirroring)	User	FR-PROF-07, FR-LOC-01–03
UC-12	Declare equipment access to get a home/bodyweight plan	User	UR-HOME-02

3. Pose coaching (FR-CV)



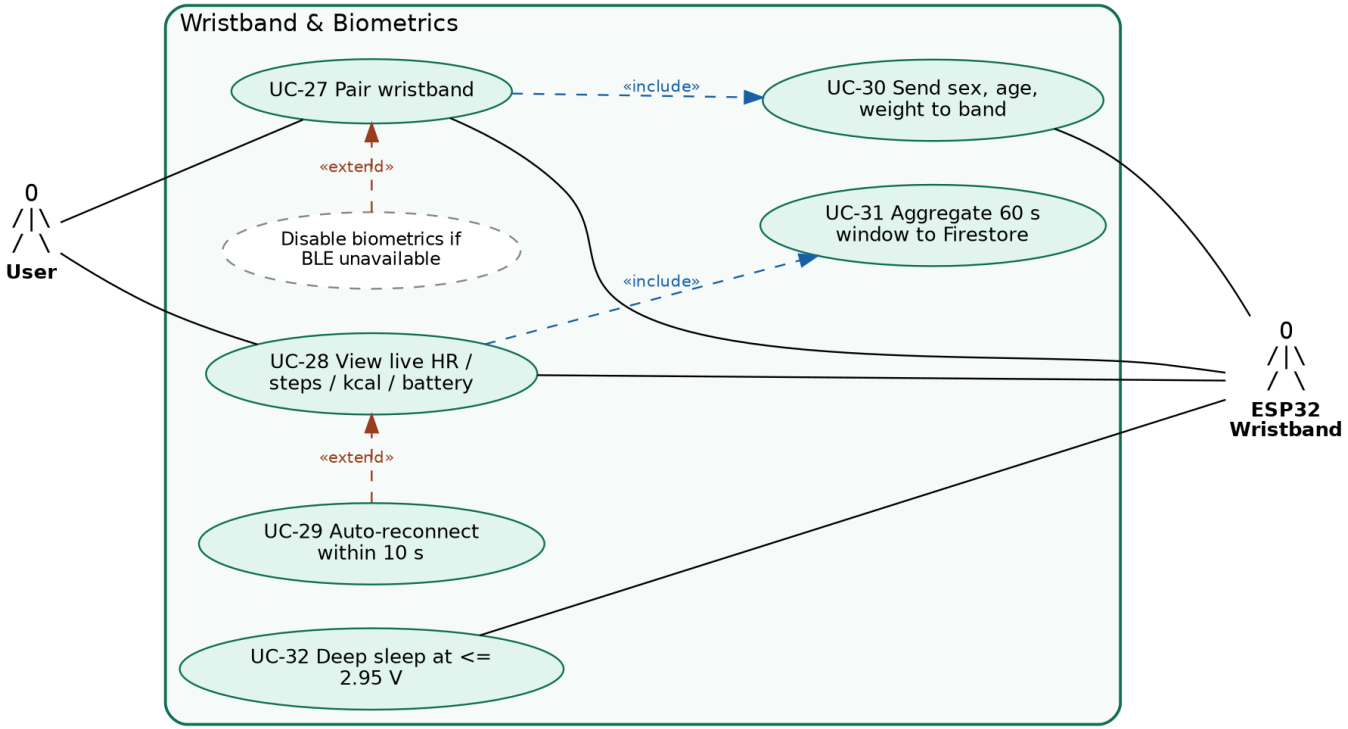
ID	Use case	Actor	Reqs
UC-13	Start a tracked set for squat / deadlift / bench press (≤ 3 taps)	User	FR-CV-01, 02, 12; NFR-USE-03
UC-14	Framing calibration with 3-second audible countdown	User	FR-CV-10
UC-15	Receive live corrective feedback (overlay + TTS, knee valgus, depth, lean…)	User	FR-CV-04, 06, 07, 08
UC-16	Automatic repetition counting with hysteresis	System	FR-CV-05, UR-EXP-01
UC-17	Suspend scoring and prompt repositioning if landmarks missing > 1 s	System	FR-CV-09
UC-18	Auto-select or manually override Lite/Full/Heavy model	User, System	FR-CV-03, NFR-COMP-03
UC-19	View per-set summary (reps, TUT, ROM, tempo, form 0–100, faults)	User	FR-CV-14
UC-20	Auto-downgrade model/resolution when FPS below floor for > 3 s	System	FR-CV-13, NFR-PWR-05

4. Workout session & safety (FR-ALRT, FR-DATA)



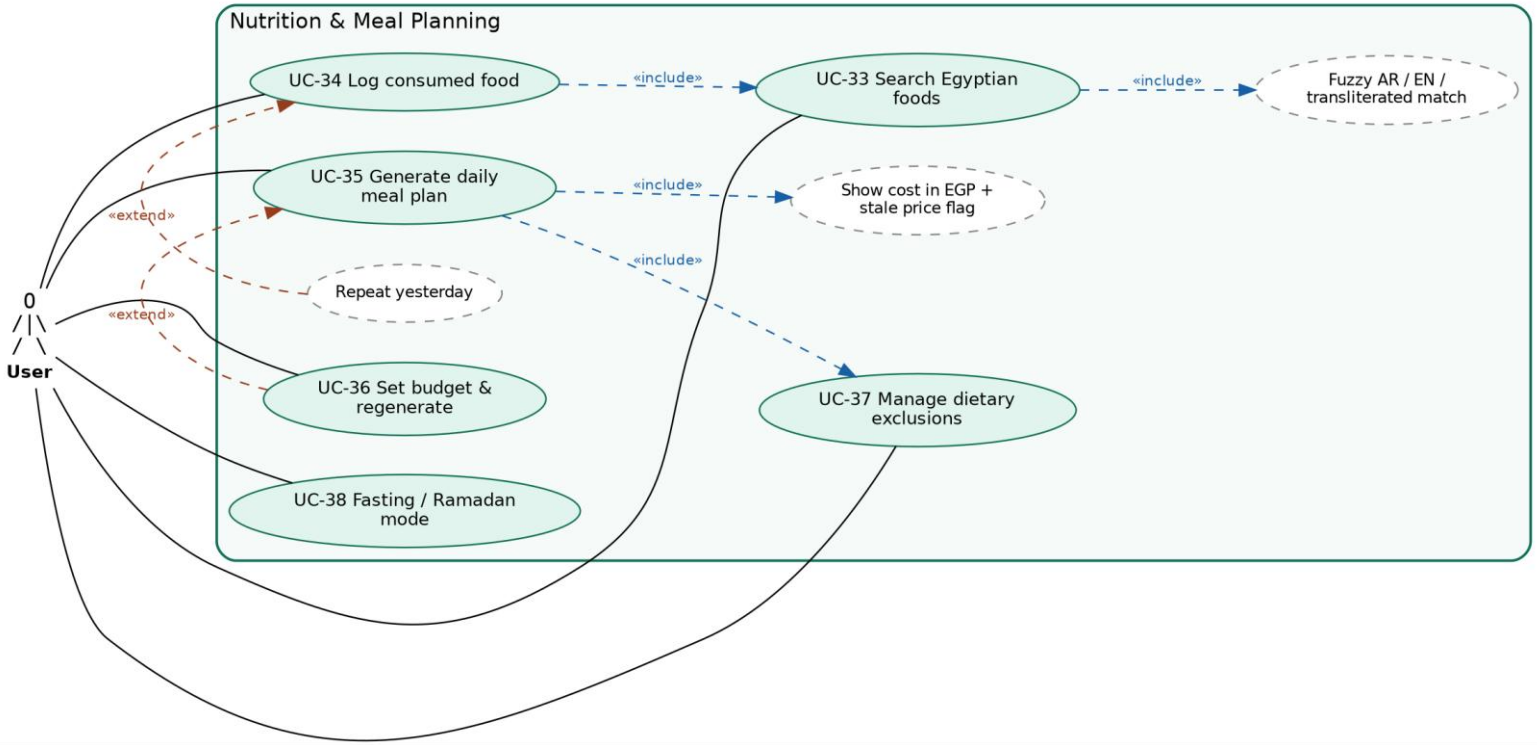
ID	Use case	Actor	Reqs
UC-21	Build an ordered workout (exercises, sets, reps, load)	User	FR-ALRT-01
UC-22	Run a workout and track per-set completion	User	FR-ALRT-01, FR-DATA-05
UC-23	Automatic inter-set rest timer	System	FR-ALRT-02
UC-24	See warm-up recommendation; acknowledge first-session safety notice	User	FR-ALRT-04, UR-NOV-02
UC-25	Get caution when HR exceeds 90% of HRmax	System	FR-ALRT-03
UC-26	Log workouts and food offline with zero loss, auto-reconcile	User	FR-DATA-03, NFR-REL-02, UR-EXP-02

5. Wristband & biometrics (BLE)



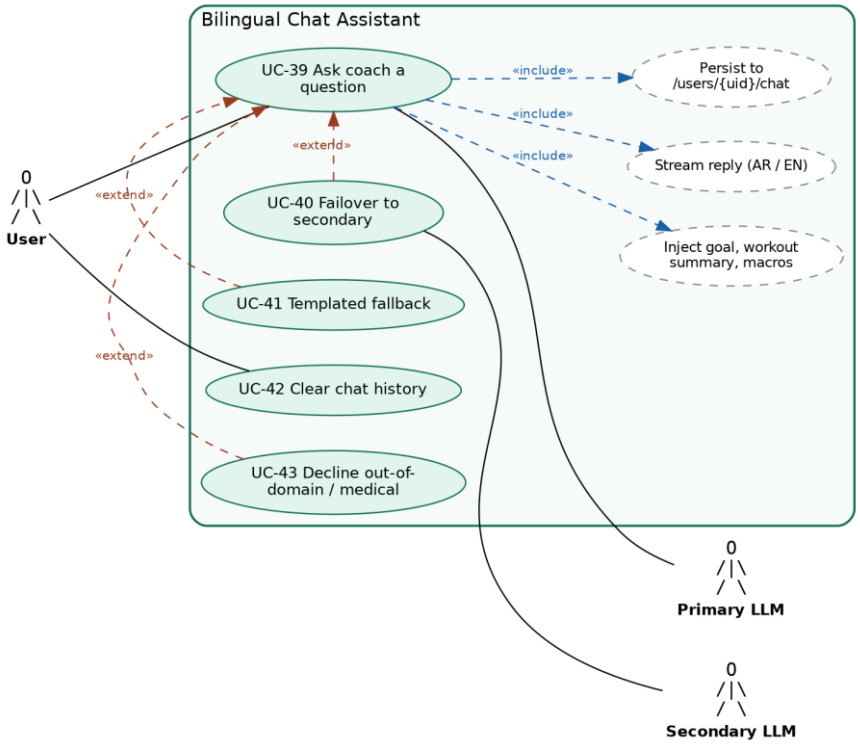
ID	Use case	Actor	Reqs
UC-27	Pair and connect to the ESP32 band over BLE	User, Wristband	C-TECH-04, NFR-COMP-04
UC-28	View live HR, calories, steps, battery (updates every 2 s, UI ≤ 200 ms)	User, Wristband	NFR-PERF-04
UC-29	Automatically re-establish link on returning to range	System	NFR-REL-04
UC-30	Send sex, age, weight to band for on-device calorie estimate	System, Wristband	Scope §1.2
UC-31	Aggregate biometric samples into one document per 60 s	System	FR-DATA-04, NFR-PERF-12
UC-32	Band enters deep sleep to protect the cell on low battery	Wristband	NFR-PWR-06

6. Nutrition & meal planning (FR-NUT)



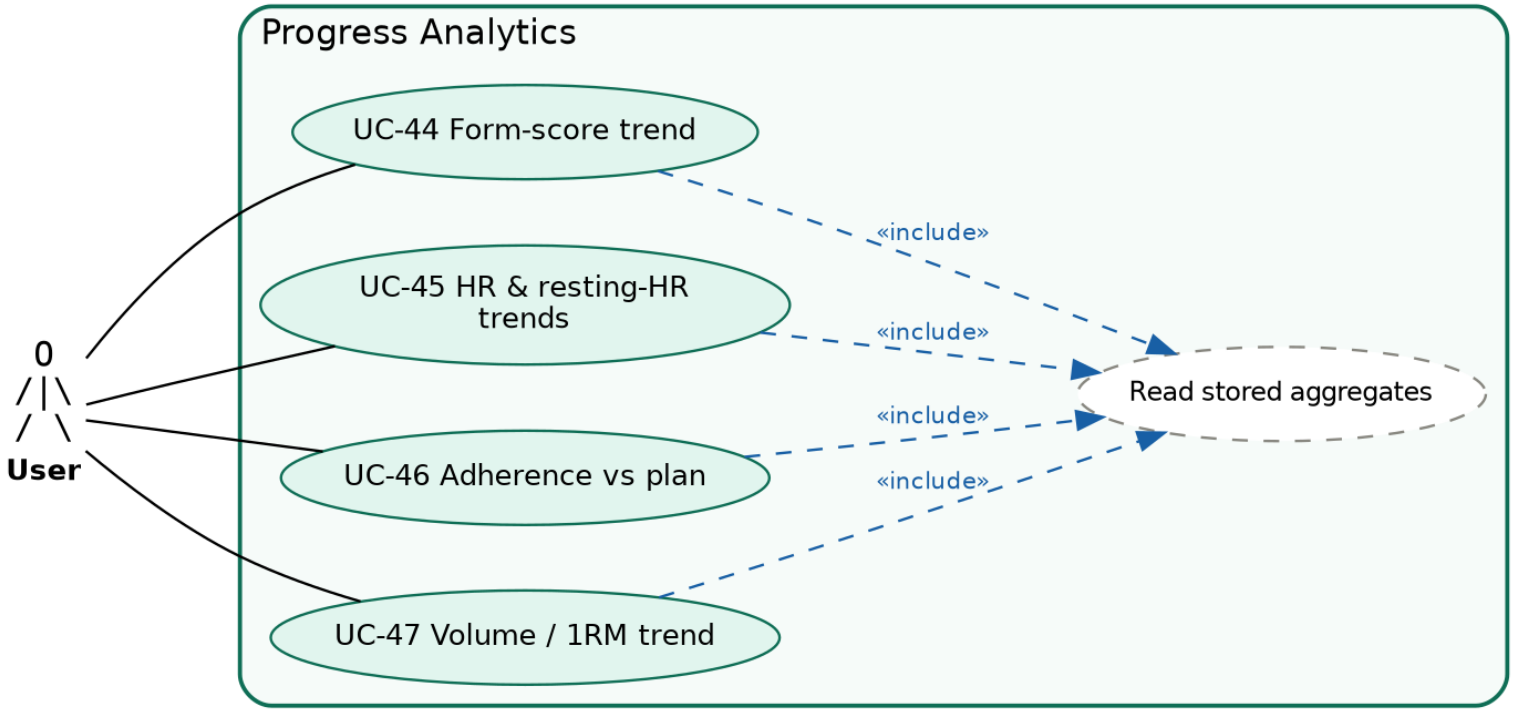
ID	Use case	Actor	Reqs
UC-33	Search foods in Arabic, English or transliteration (offline, ≤ 500 ms)	User	FR-NUT-04, FR-DATA-07, UR-NUT-01
UC-34	Log foods by search, portion or repeat-yesterday; running totals	User	FR-NUT-07
UC-35	Generate daily plan (3 meals + ≤ 2 snacks) within TDEE ±5%, protein ±10%	User	FR-NUT-05, UR-NUT-02
UC-36	Set a daily budget ceiling; view plan cost in EGP; regenerate	User	FR-NUT-03, 06
UC-37	Record exclusions; excluded foods never appear in a plan	User	FR-NUT-09, UR-NUT-03
UC-38	Fasting mode with custom eating window and Ramadan preset	User	FR-NUT-08

7. Bilingual chat assistant (FR-CHAT)



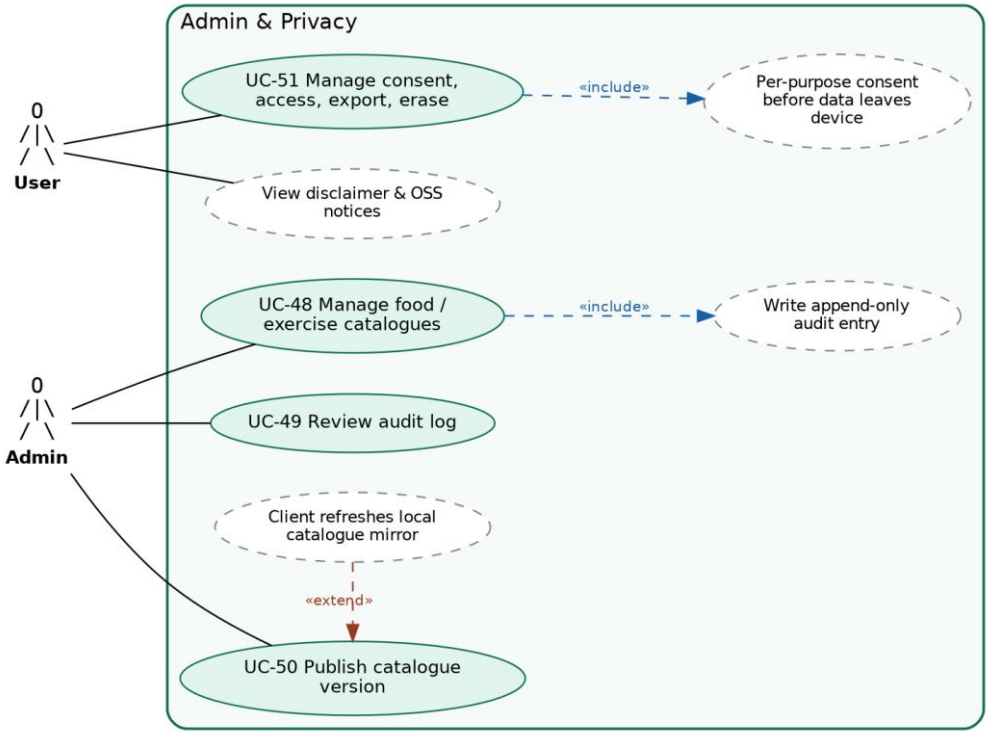
ID	Use case	Actor	Reqs
UC-39	Ask a fitness/nutrition question; streamed reply in MSA, Egyptian Arabic or English	User, LLM	FR-CHAT-01, 04, 05
UC-40	Fail over to secondary provider on error / timeout / 5 s no first token	System	FR-CHAT-02, NFR-REL-05
UC-41	Return cached/templated answer when both providers are down	System	FR-CHAT-08
UC-42	Clear stored chat history	User	FR-CHAT-07
UC-43	Politely decline out-of-domain or medical requests; show disclaimer	System	FR-CHAT-03, 06

8. Progress analytics (FR-ANL)



ID	Use case	Actor	Reqs
UC-44	Chart mean form score per exercise over time	User	FR-ANL-03
UC-45	Chart per-session HR and resting HR	User	FR-ANL-04
UC-46	Track sessions completed vs planned and minutes vs weekly target	User	FR-ANL-05
UC-47	View load / volume / 1RM trends	User	UR-EXP (no FR yet)

9. Admin & privacy (FR-DATA, NFR-SEC, NFR-LEG)



ID	Use case	Actor	Reqs
UC-48	Add/edit foods and exercises (read-all, admin-write)	Admin	FR-DATA-06, NFR-SEC-12
UC-49	Review the append-only audit log	Admin	NFR-SEC-12
UC-50	Publish catalogue version; clients refresh local mirror	Admin, System	FR-DATA-07
UC-51	Grant/withdraw consent; access, correct, erase data; read disclaimers and licences	User	NFR-LEG-01–04, NFR-SEC-06–07